

Umbrella Notes — Module 1

Computer Networks | shared core for Coal India MT (Systems), IBPS SO IT Officer (SPL-XVI), DSSSB TGT CS (DOE)

How to read this module

This is **full teaching notes** — learn directly from it. Depth is **balanced across all three exams** (middle depth by default: enough for IBPS SO, DSSSB TGT, and general PSU/Govt CS-IT-Systems exams).

■ **Green box** = how PSU/Govt exams actually test the topic (a reference lens, not a limit).

↑ **Amber box** = extra GATE-core depth needed *only* if you are targeting Coal India MT (or GATE-style PSUs). Skip for IBPS/DSSSB.

■ **Purple box** = a coverage warning (something intentionally kept light, or where an exam needs more/less).

Small sub-topics are folded under umbrella headings but never dropped. Topic-wise Q&A with explanations comes as a separate set after your walkthrough.

0 What this module covers

Computer Networks is the study of how independent devices exchange data reliably over shared media. We build it the way the layered models do — from the physical wire up to the application — because every exam (and every real protocol) is organised around that layering. The through-line: **each layer solves one class of problem and hands a clean service to the layer above.**

- **1. Foundations** — what a network is, types (LAN/MAN/WAN/PAN), topologies, transmission media, switching.
- **2. Reference models** — OSI (7 layers) and TCP/IP (4-5 layers), and how they map.
- **3. Physical & Data Link layer** — signalling basics, framing, error detection, MAC, Ethernet, devices.
- **4. Network layer** — IP addressing, IPv4, subnetting/CIDR, routing (DV, LS), ARP/DHCP/ICMP, NAT, IPv6.
- **5. Transport layer** — UDP, TCP (connection, reliability, flow & congestion control), ports, sockets.
- **6. Application layer** — DNS, HTTP/HTTPS, SMTP/POP/IMAP, FTP, and supporting services.
- **7. Security & wrap-up** — threats, encryption basics, firewalls/VPN (kept light here; expanded in the exam-wise PDFs).

■ Tested pattern & focus (PSU / Govt CS-IT-Systems)

Across IBPS SO IT, DSSSB TGT CS and most PSU/Govt CS papers, CN is a **high-frequency** scoring area — typically direct, factual, single-concept MCQs.

Heaviest-tested clusters: **OSI vs TCP/IP layer functions & which protocol sits where, IP addressing / subnetting / CIDR, device-vs-layer matching** (hub/switch/router/gateway), **TCP vs UDP**, and **port numbers**.

Numericals appear mainly in subnetting, bandwidth/data-rate, and (GATE-side) sliding-window/throughput.

1 Foundations

1.1 What a network is, and the four goals

A **computer network** is a collection of autonomous devices (hosts, or *end systems*) connected by communication links so they can exchange data and share resources. “Autonomous” matters: the machines are independent, not one controlling the others. Every design choice in networking serves four goals:

- **Delivery & correctness** — data reaches the right destination, intact.
- **Reliability** — recover from errors and failures (retransmit, reroute).
- **Performance** — measured by **bandwidth/throughput** (data per second) and **latency/delay** (time to cross).
- **Security** — confidentiality, integrity, and availability of the data in transit.

Two data-flow terms exams like to contrast: **simplex** (one direction only, e.g. keyboard→CPU), **half-duplex** (both directions, one at a time, e.g. walkie-talkie), and **full-duplex** (both directions at once, e.g. telephone).

1.2 Network types by scale

Type	Scale	Example	Owned by
PAN	~10 m (around a person)	Bluetooth earbuds, hotspot	Individual
LAN	Building / campus	Office Ethernet, college Wi-Fi	One organization
MAN	City-wide	Cable TV network, city fibre	ISP / city body
WAN	Country / global	The Internet, bank inter-branch links	Multiple / telco

LAN vs WAN is the classic contrast: LANs are privately owned, high-speed, low-delay, small area; WANs span large distances, are typically leased from carriers, and have higher delay. The **Internet** is a WAN of WANs — a network of interconnected networks.

1.3 Topologies (physical/logical arrangement)

Topology	How it connects	Strength	Weakness
Bus	All nodes on one shared backbone cable	Cheap, simple, little cabling	Backbone fault kills all; collisions
Star	All nodes to a central hub/switch	Easy to manage; node fault isolated	Central device is single point of failure
Ring	Each node to two neighbours, circular	Orderly access, no collisions (token)	One break can down the ring
Mesh	Every node linked to every other	Very robust, redundant paths	Expensive; $n(n-1)/2$ links
Tree	Hierarchy of star segments	Scalable, structured	Root/backbone dependency

■ Memory hook + formula

Mesh links for n nodes = $n(n-1)/2$ (duplex); each node needs **(n-1)** I/O ports. This exact formula is a recurring MCQ.

1.4 Transmission media

Media split into **guided** (wired) and **unguided** (wireless).

Medium	Class	Notes
Twisted pair (UTP/STP)	Guided	Cheap, common in LANs (Cat5e/6); twisting cancels noise. RJ45 connector.
Coaxial cable	Guided	Higher bandwidth than twisted pair; older Ethernet, cable TV.
Optical fibre	Guided	Light through glass; huge bandwidth, immune to EMI, long distance; costly.
Radio waves	Unguided	Omnidirectional; Wi-Fi, AM/FM; penetrate walls.
Microwaves	Unguided	Line-of-sight, directional; towers, satellites.

Medium	Class	Notes
Infrared	Unguided	Very short range, blocked by walls; old remotes.

↑ Go deeper — GATE-core (Coal India MT)

For Coal India / GATE, add the **signalling and capacity theory** that IBPS/DSSSB don't need: **Nyquist** (noiseless): $C = 2B \cdot \log_2(L)$ bits/s; **Shannon** (noisy): $C = B \cdot \log_2(1+S/N)$ bits/s; plus modulation (ASK/FSK/PSK/QAM), and delay components (transmission, propagation, queuing, processing).

1.5 Switching — how data moves through intermediate nodes

Technique	Idea	Used by
Circuit switching	Dedicated end-to-end path reserved for the whole session (like a phone call). No per-packet routing.	Traditional telephone network
Packet switching	Data split into packets, each routed independently; links shared. Efficient, resilient.	The Internet
— Datagram	Each packet routed independently; may arrive out of order (connectionless).	IP layer
— Virtual circuit	A logical path set up first; all packets follow it in order (connection-oriented).	Frame Relay, ATM, MPLS

The key contrast: circuit switching **guarantees** resources but wastes idle capacity; packet switching **shares** capacity efficiently but can suffer congestion. The Internet chose packet switching.

2 Reference Models — OSI & TCP/IP

Networks are built in **layers**: each layer performs a defined function, uses the service of the layer below, and offers a service to the layer above. This is the single most-tested area in CN, so learn it cold.

2.1 The OSI model (7 layers)

OSI (Open Systems Interconnection, by ISO) is the **conceptual reference**. Top-to-bottom: **A**pplication, **P**resentation, **S**ession, **T**ransport, **N**etwork, **D**ata Link, **P**hysical. Mnemonic (top→bottom): “*All People Seem To Need Data Processing.*”

# / Layer	Core function	Data unit	Examples
7 Application	Interface to the user's app; network services	Data / Message	HTTP, FTP, SMTP, DNS
6 Presentation	Translation, encryption, compression	Data	SSL/TLS, JPEG, ASCII
5 Session	Establish, manage, sync dialogues	Data	NetBIOS, RPC
4 Transport	End-to-end delivery, reliability, flow/congestion	Segment	TCP, UDP
3 Network	Logical addressing & routing across networks	Packet	IP, ICMP, routers
2 Data Link	Node-to-node framing, MAC, error detection	Frame	Ethernet, switches, MAC
1 Physical	Bits onto the medium; signalling, connectors	Bit	cables, hubs, repeaters

■ Two hooks exams reward

PDU by layer: Data (5-7) → Segment (4) → Packet (3) → Frame (2) → Bit (1).

Encapsulation: as data goes down the stack each layer adds its own **header** (Transport→segment, Network→packet, etc.); the receiver strips them going up. “Headers added top-down, removed bottom-up.”

2.2 The TCP/IP model (the real Internet stack)

TCP/IP is the **practical** model the Internet actually runs on. It has 4 layers (sometimes drawn as 5 by splitting the bottom). It predates OSI and is protocol-driven rather than theoretical.

TCP/IP layer	Maps to OSI	Key protocols
Application	OSI 5+6+7	HTTP, FTP, SMTP, DNS, DHCP
Transport	OSI 4	TCP, UDP
Internet	OSI 3	IP, ICMP, ARP
Network Access / Link	OSI 1+2	Ethernet, Wi-Fi, ARP

■ OSI vs TCP/IP — the comparison they love

OSI = 7 layers, **conceptual**, protocol-independent, defined before protocols. TCP/IP = 4 layers, **practical**, built around existing protocols.

OSI separates **Session** and **Presentation**; TCP/IP merges them into Application. OSI's Physical+Data Link = TCP/IP's Network Access.

Both are layered; both use encapsulation. TCP/IP is what the Internet uses.

■ Coverage note / warning

Session & Presentation layer details (dialog control, translation/encryption specifics) are kept light here — they're low-frequency for IBPS/DSSSB. If you meet them in a mock, the one-line functions in the table above are sufficient.

3 Physical & Data Link Layer + Devices

3.1 Physical layer (in brief)

The physical layer moves raw **bits** across the medium. Its concerns: encoding bits into signals, data rate, synchronization, and physical topology/connectors. Key terms: **bandwidth** (range of frequencies, or loosely the max data rate), **bit rate** (bits/s), **baud rate** (signal changes/s), and **bit rate = baud rate × bits per signal**.

3.2 Data Link layer — turning a raw link into a reliable hop

The data link layer (DLL) delivers data **node-to-node** (one hop) and has two sublayers: **LLC** (Logical Link Control — flow/error control interface) and **MAC** (Media Access Control — who gets the shared medium). Its jobs:

- **Framing** — wrap bits into **frames** with a header/trailer so the receiver knows where each frame starts and ends.
- **Physical (MAC) addressing** — the 48-bit **MAC address** burned into the NIC identifies the node on the local link.
- **Error detection** — catch corrupted frames (below).
- **Flow control** — stop a fast sender overwhelming a slow receiver.
- **Media access control** — arbitrate the shared medium (below).

3.3 Error detection

Method	Idea	Strength
Parity bit	Add 1 bit to make count of 1s even/odd	Detects single-bit errors only
Checksum	Sum data words, send complement; recompute at RX	Used in IP/TCP/UDP headers
CRC	Treat data as polynomial, divide by generator; send remainder	Very strong; standard in Ethernet

Detection vs correction: the above *detect* errors (then request retransmission). **Hamming code** can *correct* single-bit errors by adding redundant parity bits at power-of-two positions.

3.4 Media Access Control (who transmits when)

- **CSMA/CD** (Carrier Sense Multiple Access / Collision Detection) — classic **wired Ethernet**: listen before sending; if a collision is detected, back off and retry.
- **CSMA/CA** (Collision Avoidance) — **Wi-Fi**: collisions can't be detected on radio, so avoid them (wait + acknowledgements).
- **Token passing** — a token circulates; only its holder may transmit (Token Ring). No collisions.

3.5 Networking devices — match device to layer

Device	Layer	What it does
Repeater	1 Physical	Regenerates/amplifies the signal to extend distance
Hub	1 Physical	Multiport repeater; broadcasts to all ports (one collision domain) — obsolete
Bridge	2 Data Link	Connects two LAN segments; filters by MAC address
Switch	2 Data Link	Multiport bridge; forwards frames only to the target MAC (separate collision domains)
Router	3 Network	Connects different networks; forwards packets by IP; separates broadcast domains
Gateway	Up to 7	Protocol translator between dissimilar networks/architectures
NIC	1-2	Network interface card; holds the MAC address

■ Tested pattern & focus (PSU / Govt CS-IT-Systems)

Device-to-layer matching is one of the most repeated CN MCQs in IBPS/SSC/PSU papers. Nail: hub/repeater = L1, switch/bridge = L2, router = L3, gateway = up to L7.

Follow-up favourites: *which device separates collision domains? (switch/bridge/router) vs broadcast domains? (router).*

4 Network Layer — IP, Addressing, Routing

The network layer moves **packets** from source host to destination host across multiple networks. Two jobs: **logical addressing** (IP) and **routing** (choosing the path).

4.1 IPv4 addressing

An **IPv4 address** is **32 bits**, written as four dotted decimal octets (e.g. 192.168.1.10), each 0-255. It has a **network part** and a **host part**. Total space $\approx 2^{32} \approx 4.3$ billion addresses — now exhausted, which is why NAT and IPv6 exist.

Class	First octet range	Default mask	Use
A	0 - 127	255.0.0.0 (/8)	Very large networks
B	128 - 191	255.255.0.0 (/16)	Medium networks
C	192 - 223	255.255.255.0 (/24)	Small networks
D	224 - 239	—	Multicast
E	240 - 255	—	Experimental / reserved

■ Reserved ranges worth memorising

Private (RFC 1918): 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16. **Loopback**: 127.0.0.1. **APIPA**: 169.254.x.x. These are frequent MCQs.

4.2 Subnetting & CIDR

Subnetting splits one network into smaller subnets by borrowing host bits for the network part. **CIDR** (Classless Inter-Domain Routing) drops rigid classes and uses a **/n** prefix meaning 'n network bits'. The key working facts:

- **Hosts per subnet** = $2^{(32-n)} - 2$ (subtract 2 for the network address and broadcast address).
- **Subnet mask**: /24 = 255.255.255.0, /25 = 255.255.255.128, /26 = 255.255.255.192, and so on.
- **Number of subnets** when borrowing b bits = 2^b .
- Worked example: a /26 has $32-26 = 6$ host bits $\rightarrow 2^6 - 2 = 62$ **usable hosts** per subnet, and a /24 split into /26 gives $2^2 = 4$ **subnets**.

■ Tested pattern & focus (PSU / Govt CS-IT-Systems)

Subnetting is the #1 numerical in CN across banking/PSU/SSC. Practise: given an IP + mask, find network address, broadcast address, usable host range, and host count.

Learn the mask $\leftrightarrow$ prefix table (/24–/30) by heart; most questions resolve in seconds once you know it.

4.3 Support protocols (ARP, DHCP, ICMP, NAT)

Protocol	Job	One-liner
ARP	IP $\rightarrow$ MAC	Resolves a known IP to its MAC on the local link
RARP	MAC $\rightarrow$ IP	Legacy reverse of ARP
DHCP	Auto-config	Dynamically leases IP, mask, gateway, DNS to hosts
ICMP	Diagnostics	Error/echo messages; powers ping and traceroute
NAT	Address sharing	Maps many private IPs to one/few public IPs — eases IPv4 shortage

4.4 Routing

Routing is choosing paths for packets. Two big algorithm families:

- **Distance Vector (DV)** — each router shares its whole routing table with neighbours; simple but slow to converge ('count-to-infinity'). Example: **RIP**.

- **Link State (LS)** — each router floods link info; every router builds the full map and runs **Dijkstra's shortest path**. Faster convergence. Example: **OSPF**.
- **Static vs dynamic** — static routes are hand-configured; dynamic routes adapt via protocols (RIP/OSPF/BGP). **BGP** routes between autonomous systems (the Internet's backbone protocol).

↑ **Go deeper — GATE-core (Coal India MT)**

Coal India / GATE go further: **Dijkstra** and **Bellman-Ford** step-by-step, count-to-infinity + split-horizon fixes, hierarchical routing, and IP header field details (TTL, fragmentation offset, MTU/fragmentation math). IBPS/DSSSB need only the DV-vs-LS contrast and example protocols above.

4.5 IPv6 (in brief)

IPv6 uses **128-bit** addresses (written as eight hex groups) to solve IPv4 exhaustion — a practically unlimited space. It also brings simpler headers, built-in security (IPsec), and no need for NAT. For IBPS/DSSSB, remember: **IPv4 = 32-bit**, **IPv6 = 128-bit**, and IPv6 exists because IPv4 ran out.

5 Transport Layer — TCP & UDP

The transport layer gives **end-to-end** (process-to-process) delivery. It adds **port numbers** to identify the application, and — in TCP's case — reliability on top of the unreliable network layer.

5.1 Ports & sockets

A **port** (16-bit, 0-65535) identifies a specific application/service on a host. A **socket** = IP address + port, the unique endpoint of a connection. Well-known ports to memorise:

Port	Service	Port	Service
20/21	FTP (data/control)	53	DNS
22	SSH	80	HTTP
23	Telnet	110	POP3
25	SMTP	143	IMAP
443	HTTPS	67/68	DHCP

■ Tested pattern & focus (PSU / Govt CS-IT-Systems)

Port numbers are near-guaranteed MCQs. Memorise the table above; HTTP=80, HTTPS=443, DNS=53, SMTP=25, FTP=20/21, SSH=22 are the most asked.

5.2 TCP vs UDP — the central contrast

Feature	TCP	UDP
Connection	Connection-oriented (handshake first)	Connectionless (just send)
Reliability	Reliable — ACKs, retransmission, ordering	Unreliable — no guarantee, no reorder
Speed / overhead	Slower, heavier header (20 bytes)	Faster, light header (8 bytes)
Flow/congestion ctrl	Yes	No
Use cases	Web, email, file transfer	Streaming, VoIP, DNS, online games

TCP three-way handshake (connection setup): **SYN** → **SYN-ACK** → **ACK**. Connection teardown uses FIN/ACK exchanges. This handshake sequence is a very common question.

5.3 Flow control & congestion control

- **Flow control** — protects the *receiver* from being overwhelmed. TCP uses a **sliding window** (receiver advertises how much it can accept).
- **Congestion control** — protects the *network* from overload. TCP uses **slow start, congestion avoidance, fast retransmit/recovery**.
- Simple ARQ protocols that underpin the idea: **Stop-and-Wait, Go-Back-N, Selective Repeat**.

↑ Go deeper — GATE-core (Coal India MT)

Coal India / GATE expect the **numericals**: sliding-window efficiency, throughput = window / RTT, Go-Back-N vs Selective-Repeat window sizes ($2^m - 1$ vs $2^{(m-1)}$), and TCP congestion-window evolution graphs. IBPS/DSSSB only need the definitions and which layer/protocol does what.

6 Application Layer

The application layer is where network-using programs live. You mostly need **what each protocol does, its transport, and its port**.

Protocol	Purpose	Transport / Port
DNS	Resolves domain names → IP addresses (the Internet's phonebook)	UDP (TCP for big)/53
HTTP / HTTPS	Web page transfer; HTTPS = HTTP over TLS (encrypted)	TCP / 80, 443
SMTP	Sending email between servers	TCP / 25
POP3 / IMAP	Retrieving email to a client (POP downloads; IMAP syncs)	TCP / 110, 143
FTP	File transfer (separate control + data channels)	TCP / 21, 20
Telnet / SSH	Remote login (SSH = encrypted)	TCP / 23, 22
DHCP	Auto IP configuration	UDP / 67, 68

DNS resolution flow (commonly asked): the resolver queries a **root** server → **TLD** server (.com) → **authoritative** server for the domain, caching results along the way. **URL** = protocol + domain + path; the domain is resolved by DNS before any HTTP request is sent.

■ Tested pattern & focus (PSU / Govt CS-IT-Systems)

Application layer questions are almost always **protocol→purpose→port** matching, plus **DNS basics** and **HTTP vs HTTPS**. High yield, low effort.

7 Security — essentials only

Kept deliberately light here (it expands in the IBPS and DSSSB exam-wise PDFs, where cyber security is its own tested area). The must-know minimum:

- **CIA triad** — Confidentiality, Integrity, Availability: the three security goals.
- **Symmetric encryption** (one shared key, e.g. AES — fast) vs **asymmetric** (public/private key pair, e.g. RSA — used for key exchange, digital signatures).
- **Firewall** — filters traffic by rules at the network boundary. **VPN** — encrypted tunnel over a public network.
- **Common threats** — malware, phishing, DoS/DDoS, man-in-the-middle, spoofing.

■ Coverage note / warning

This section is a summary, by design. **Full cyber-security depth (cryptography, attacks, digital signatures, PKI) lives in the exam-wise PDFs** — it is a distinct, more heavily-tested area for IBPS SO IT and part of DSSSB DOE (Computer Network Security). Don't treat this box as complete for those exams.

Rapid recap — highest-yield facts

Area	Nail these
Layers	OSI 7 (APSTNDP), TCP/IP 4; PDU order Data-Segment-Packet-Frame-Bit; encapsulation adds headers top-down
Devices	Hub/Repeater=L1, Switch/Bridge=L2, Router=L3, Gateway=L7; router separates broadcast domains
Addressing	IPv4=32-bit, classes A/B/C, private ranges, hosts= $2^{(32-n)}-2$, mask $\leftrightarrow$ prefix table
Support	ARP=IP $\rightarrow$ MAC, DHCP=auto-config, ICMP=ping, NAT=private $\rightarrow$ public
Routing	DV(RIP) vs LS(OSPF, Dijkstra); BGP between AS
Transport	TCP reliable/connection-oriented vs UDP fast/connectionless; 3-way handshake SYN/SYN-ACK/ACK
Ports	HTTP 80, HTTPS 443, DNS 53, SMTP 25, FTP 20/21, SSH 22
Application	DNS name $\rightarrow$ IP; HTTP/HTTPS; SMTP/POP/IMAP; FTP

✓ Before your walkthrough

You now have the full shared-core CN picture at balanced depth. In our walkthrough we'll pinpoint anything unclear, then I'll hand you the **topic-wise Q&A set** (real PSU/Govt exam questions with explanations) so you can take live practice.

If you're targeting Coal India, work through the amber ↑ **GATE-core** boxes as a second pass — that's the only extra depth CN needs beyond this document.